

Inferred Transformations for Structure-Compatible Generalization

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Phase 2: supported discovery plus compatibility-guided training

Abstract

Phase one showed that oracle transformation compatibility predicts OOD behavior under underspecification. This phase weakens the oracle: candidate transformations are admitted only when observed train-label overlaps support the induced input and label action, then the discovered family is used as a train-time compatibility regularizer. The result is a finite-domain test of OOD prediction and control without using OOD labels for selection or training.

1. Regime Transition

The old regime supplied the deployment group. The new regime infers supported modular shifts from training evidence, rejects vacuous non-identity transformations, and applies compatibility pressure to unlabeled finite-domain inputs.

Field	Value
GPU	L4
configs	180
epochs	450
regularization values	[0.0, 0.05, 0.2]
rows	542

Table 1. Phase-two Modal L4 manifest.

2. Predictor Correlations

Predictor	Pearson r	Spearman r	N
compatibility_true	0.862	0.394	540
negative_train_loss	0.454	0.266	540
compatibility_discovered	0.413	0.389	540
compatibility_inferred	0.413	0.389	540
id_validation_accuracy	0.379	0.162	540
train_accuracy	0.379	0.162	540
compatibility_wrong	0.097	0.096	540

Table 2. Phase-two predictors against OOD.

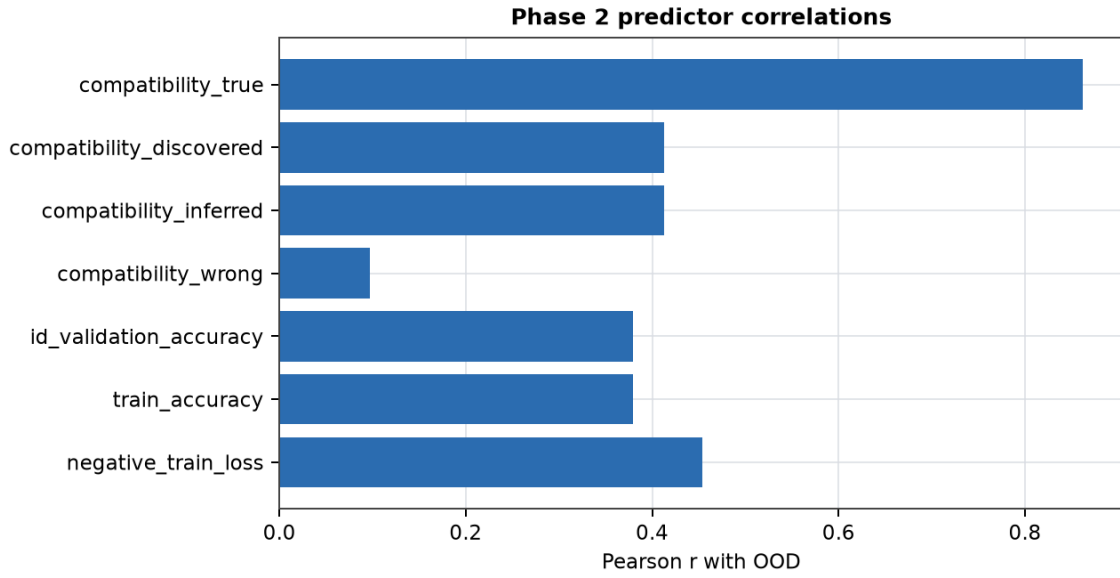


Figure 1. Discovered compatibility is compared with oracle, wrong-group, and standard training proxies.

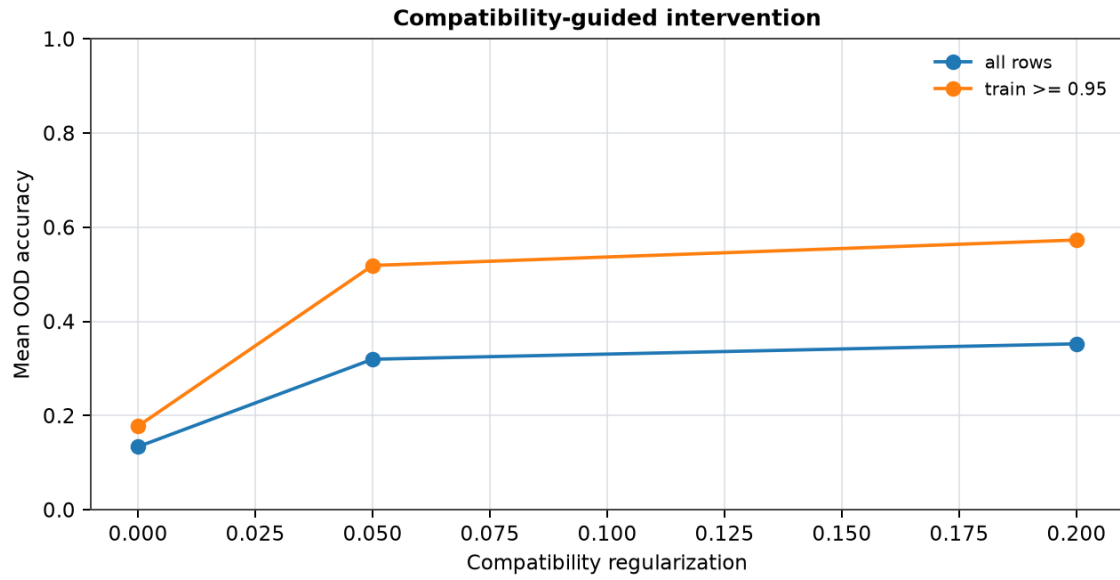


Figure 2. Compatibility-guided regularization is evaluated without OOD labels; OOD is measured only after training.

3. Intervention

Reg	N	Mean train	Mean OOD	High-ID N	High-ID OOD
0.000	180	0.719	0.134	98	0.178
0.050	180	0.719	0.320	98	0.519
0.200	180	0.723	0.352	99	0.573

Table 3. Regularization arms.

4. Scope

This is a finite modular-domain result. It supports the broader OOD-certifiability-lite program by showing the first discovery plus intervention path, but it does not yet solve learned transformation generation for vision or language.

The next operation is to make the transformation generator learned rather than enumerated, then carry the same protocol to rotations and paraphrase/template substitutions.